Software Design Document (SDD)

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CSE General

1. Introduction

The E-Book Management System is a web-based application that enables users to sign up, upload e-books, browse books by categories, and read books uploaded by other users and also download it. This document provides the detailed architecture about the software system of E-book Management System.

2. System Architecture

I. **Presentation Layer**
- User Interface (Web Application)
- Navbar with categories, login/signup
- Supports Light/Dark mode for better user experience
- Displays books and reading interface

II. **Business Logic Layer**
- Handles application logic
- User authentication and authorization
- Book upload and access control
- User shall read books online and can also download them

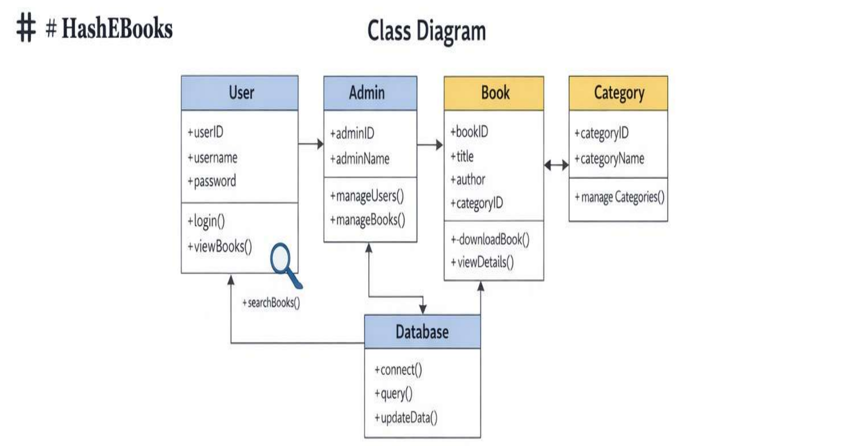
III. **Data Layer**
- Stores user data and book metadata
- Manages book storage
- Supports cloud or local database

IV. **Integration Layer**
- Connects frontend with backend services
- Handles API communication
- Integrates authentication and storage services

3. Detailed Design

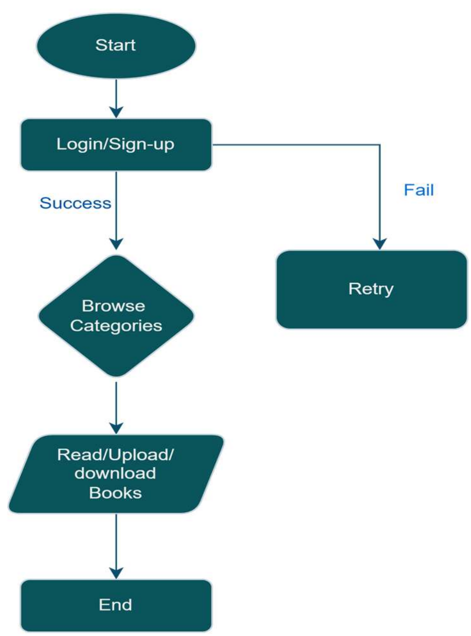
Class diagram

- The class diagram represents the static structure of the system, showing classes such as **User** and **Book** along with their attributes and relationships.
- It helps in understanding how objects are created and how different classes interact within the system.



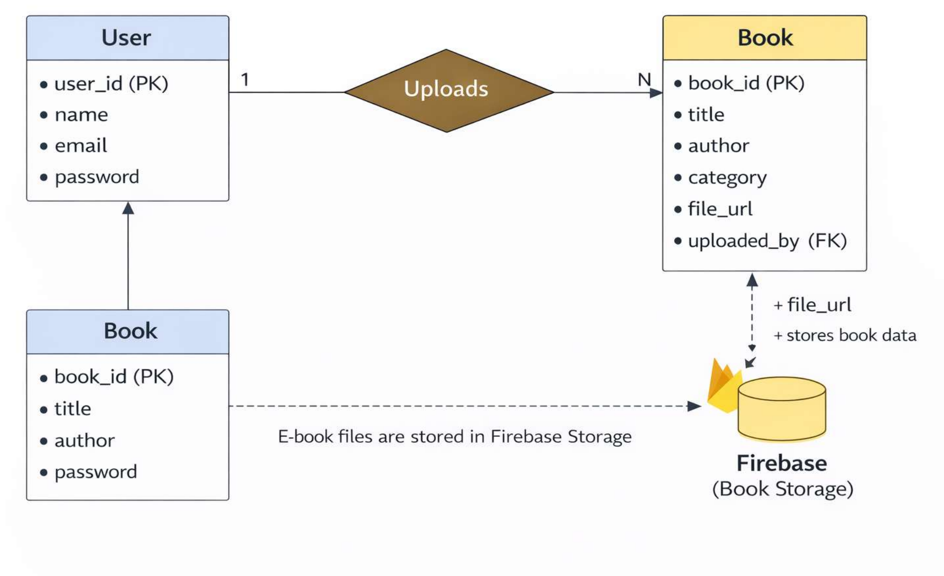
Data Flow diagram

- The data flow diagram illustrates how data moves between users, the E-Book Management System, and Firebase storage.
- It helps identify input, processing, and output of the system, ensuring smooth data communication.



4. Database Design

- The design diagram explains the internal architecture of the system, including module structure, logic flow, and data handling.
- It provides a blueprint for developers to implement the system efficiently and maintain consistency.



5. Design Constraints

Hardware Constraints

- The system must run on standard computers and mobile devices without requiring high-end hardware.
- Performance should remain acceptable even on devices with limited memory and processing power.

Software Constraints

- The application depends on specific software technologies such as web browsers, Firebase services, and supported operating systems.
- Compatibility must be maintained across different devices and platforms.

Regulatory Constraints

- The system must comply with copyright laws related to uploading and sharing digital books.
- User data must be handled securely in accordance with data protection and privacy policies.